

Enhancing accuracy and interpretability in EEG-based medical decision making using an explainable ensemble learning framework application for stroke prediction

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ABSTRACT

Medical decision making increasingly relies on machine learning algorithms to analyze complex patient data and provide recommendations. However, the lack of interpretability in “black box” models has limited their adoption in clinical practice, which demands transparency and justification. Echo State Networks (ESNs), a recurrent neural network architecture, have shown promise for medical Decision Support Systems (DSS) applications due to their accuracy in modeling time series data, such as physiological signals. However, conventional ESNs suffer from a lack of interpretability, which limits their usefulness for clinical decisions. This study proposes a multi-level framework to optimize both accuracy and interpretability for improved medical decision support. The framework is specifically designed for EEG data, which represents a non-invasive and continuous recording of brain activity. EEG data is ideal for recording information about the internal state of the brain, which is not always translated by perceptible external manifestations. The framework consists of four components: (1) data preprocessing and optimized feature selection utilizing a filter-based feature importance algorithm, (2) ensemble learning via Ensemble ESNs (*E-ESNs*) that combines the predictions of multiple ESNs to reduce variance and improve accuracy, (3) Global and Local model explanation techniques to allow users to understand the decision-making process and identify the most influential features for each prediction, and (4) decision-making for presenting diagnoses and recommendations. The framework was evaluated on an EEG dataset for stroke prediction, a valuable use case for informed clinical decisions and resource allocation. The results showed that the framework significantly outperformed baseline related works with an accuracy of 96.5% and provides insights into the *E-ESN* model's predictions. This transparency enhances trust and comprehension among healthcare practitioners. These findings suggest that the proposed framework can be a valuable tool for medical DSS applications.

1. Introduction

In the realm of modern healthcare, medical decision-making has been significantly enhanced through the adoption of invaluable tools known as medical Decision Support Systems (DSS) [25]. These systems empower clinicians to deliver precise and well-informed patient care [30]. Several medical DSS have embraced the utilization of Electroencephalogram (EEG) data as a valuable resource [26]. The rich information contained within EEG signals has made it a compelling data source for various medical applications. From brain-computer interfaces to stroke prediction and cognitive assessment, EEG data has shown remarkable potential in enhancing the accuracy and efficiency of

medical decision-making [8]. Leveraging advanced computational techniques and machine learning algorithms, these DSS extract meaningful insights from EEG recordings, aiding healthcare professionals in making well-informed and timely decisions for improved patient care and outcomes. The integration of EEG data in medical DSS reflects the ongoing quest for innovative and effective tools to support clinical practices and elevate healthcare standards [26].

EEG data analysis presents unique challenges due to its intricate and variable nature. EEG recordings are susceptible to substantial amounts of noise originating from diverse sources, including eye movements, muscle activity, and electrical interference [18,66]. Moreover, the data exhibits continuous wave-like patterns that necessitate the utilization of

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sophisticated signal processing techniques to extract meaningful features [7]. A significant hurdle in EEG preprocessing involves disentangling brain activity signals from noise and other artifacts. Another complexity lies in capturing the temporal dynamics within EEG data, as the signals encompass crucial information about rhythmic electrical activity in the brain over time [5]. Time series algorithms capable of modeling sequential dependencies prove invaluable in connecting EEG signals recorded at different time points. Promising approaches, such as Long Short-Term Memory networks and Echo State Networks, have demonstrated their efficacy in analyzing EEG connections over short time intervals to comprehend changes in brain states [68].

While previous research efforts have certainly advanced the field of classifying EEG data through the application of machine learning techniques, a critical gap persists in the development of interpretable models that can effectively support clinical decision-making [40]. In the context of this study, we are embarking on a mission to bridge this gap by not only focusing on accurate classification but also placing a significant emphasis on model explainability [69]. This research work is not just about advancing the state-of-the-art in EEG data classification; it is about creating a paradigm shift in the field by delivering models that are not only highly accurate but also highly interpretable, ultimately benefiting both patients and clinicians alike.

In recent years, there has been growing interest in using Echo State Networks (ESNs) for analyzing EEG data in the context of brain-computer interfaces (BCIs) and other applications [4,15,16]. These studies have demonstrated that ESNs can achieve high accuracy in classifying EEG signals, even in the presence of noise and other sources of variability. One potential advantage of ESNs is that they can handle high-dimensional data with complex temporal dynamics, which is characteristic of EEG signals. However, more research is needed to fully explore the potential of ESNs for EEG data analysis and to compare their performance to other machine learning techniques. Another challenge to be addressed in this paper is the need for interpretable models. In fact, EEG-based medical DSS need to be explainable to ensure that their predictions can be understood by clinicians. In this context, we propose a multi-level framework for EEG-based Medical DSS that would be a valuable tool for developing new diagnostic tools. This framework consists of four components: (1) a data preprocessing and optimized feature selection module utilizing a filter-based feature importance algorithm, (2) an ensemble learning via a proposed Ensemble ESNs (E-ESNs) that combines the predictions of multiple ESNs to reduce variance and improve accuracy, (3) Model explanation using Artificial Intelligence Explainability techniques to allow users to understand the decision-making process and identify the most influential features for each prediction, and (4) decision-making for presenting diagnoses and recommendations.

To evaluate our framework, we apply it for EEG-based stroke prediction. In fact, stroke prediction is a critical research area aiming to identify at-risk individuals early so timely interventions can reduce the devastating impact of this event [27,52]. However, the complex and multifaceted nature of stroke makes accurate prediction challenging. Recent advances in machine learning and data analysis have shown potential for improving stroke prediction accuracy. Extensive research on stroke diseases is actively being conducted, driven by the growing aging population. Stroke, a life-threatening condition among the elderly, necessitates continuous medical observation and monitoring due to its high recurrence and mortality rates [67]. Previous studies have primarily focused on the classification of stroke using medical images [48]. However, this clinical approach relying on imaging techniques is associated with drawbacks such as high cost, time consumption, and the requirement for bulky equipment. In recent years, there has been a rising interest in utilizing non-invasive EEG measurements as an alternative to overcome these limitations [2]. EEG readily detects deviations in brain rhythms from stroke, making it a valuable diagnostic tool for cognitive assessment [29]. Moreover, EEG signals can serve as invaluable resources for real-time physiological monitoring and early prognosis in

daily life [44]. Numerous studies have employed EEG to predict stroke outcomes in medical and healthcare settings. Machine learning algorithms, such as support vector machines (SVMs), random forests, and neural networks, have been used to classify EEG data from stroke patients and predict stroke occurrence or outcome [63]. While there has not been much research specifically on the use of ESNs for stroke prediction, our framework may be a promising approach for analyzing EEG data in the context of stroke.

The remainder of this paper is organized as follows. Section 2 covers the theoretical foundations of our research, namely EEG-based medical Decision Support Systems, Echo State Networks and Explainable Artificial Intelligence. Section 3 introduces the proposed multi-level framework. Section 4 provides the case study concerning EEG-based stroke prediction. Section 5 discusses the proposed and applied framework. Finally, Section 6 presents the conclusion and suggestions for future work.

2. Theoretical background

2.1. EEG-based medical decision support systems

Medical decision making is increasingly being supported by computerized Decision Support Systems (DSS). Electroencephalography (EEG) based DSS can help analyze brain signals and support clinical diagnosis and treatment decisions [26]. EEG offers an accessible, noninvasive way to measure brain activity and has been used in DSS for various medical applications. EEG signals have valuable applications in the diagnosis and treatment of medical conditions through multiple avenues [14]. Firstly, abnormalities observed in EEG patterns can serve as indicators of neurological disorders. Specific brain wave patterns are associated with particular medical conditions; for instance, epileptic seizures generate distinctive spikes and sharp waves in EEG readings, while strokes can lead to a deceleration of brain rhythms. By detecting and analyzing these abnormalities, clinicians can effectively diagnose various neurological disorders. Secondly, changes in EEG patterns over time can be utilized to track disease progression and evaluate the effectiveness of treatments.

By conducting repeated EEG recordings, medical professionals can gain insights into how a patient's condition is evolving and whether the implemented treatments are yielding positive results. For example, a reduction in epileptic spikes following medication administration may signify improved seizure control. Moreover, EEG features can be employed as biomarkers for medical conditions. Quantitative measures such as power in specific frequency bands, synchronization between different regions of the brain, and complexity metrics can function as objective indicators of a patient's state. These features enable the classification of patients compared to control groups and the differentiation of subtypes within a particular condition [26]. In addition, with advancements in neuroprosthetics, brain-computer interfaces utilizing EEG signals hold the potential to restore lost functions. In the future, it may be possible to directly control devices that compensate for motor, sensory, or cognitive deficits caused by disease or injury through the analysis of EEG signals. Therefore, EEG signals offer a noninvasive means of observing brain function. Through the analysis of EEG patterns, features, and signals, valuable information can be obtained to aid in the diagnosis of various medical conditions, monitor their progression, and potentially restore lost functions through technological advancements [26].

Data preprocessing plays a vital role in EEG data processing, as it is crucial for enhancing the quality and reliability of the data before analysis. It involves various steps such as noise reduction, artifact removal, and signal normalization, which are essential for obtaining accurate and meaningful results. The significance of data preprocessing lies in its ability to mitigate unwanted distortions and artifacts present in EEG signals, thereby improving the interpretability and validity of the subsequent analysis [47]. However, EEG data processing presents

several challenges that need to be addressed. One major challenge is the presence of various artifacts, including eye blinks, muscle activity, and electrical interference, which can significantly corrupt the EEG signals. Detecting and removing these artifacts while preserving the integrity of the underlying brain activity remains a complex task. Researchers have been actively working on developing sophisticated algorithms and techniques to effectively identify and eliminate these artifacts, ensuring that the processed EEG data accurately represents the underlying neural activity [47]. Moreover, the non-stationary nature of EEG signals poses another challenge in their processing. EEG signals exhibit temporal dynamics and variations, making it necessary to consider time-varying properties during analysis. Traditional signal processing methods may not adequately capture these temporal variations, requiring the utilization of advanced techniques such as time-frequency analysis and adaptive filtering [55,58]. Incorporating these approaches effectively addresses the non-stationarity of EEG signals and enables a more comprehensive analysis of their dynamic characteristics [47]. In recent years, additional research efforts have focused on integrating machine learning and deep learning techniques into EEG data preprocessing. These approaches have shown great promise in automating artifact detection, feature extraction, and classification tasks, thereby streamlining the data preprocessing pipeline. By leveraging the power of these advanced algorithms, researchers have achieved significant advancements in EEG data processing, leading to improved accuracy, efficiency, and reliability in various applications, including clinical diagnosis, brain-computer interfaces, and cognitive neuroscience [58].

Machine learning techniques have revolutionized the analysis of EEG data and the development of clinical decision support systems [22]. These techniques enable the creation of automated classifiers, predictors, and decision-making models based on quantitative EEG features and signals [22]. The main objective is to build accurate and robust models that assist clinicians in making informed diagnoses and treatment decisions. Machine learning classifiers trained on relevant EEG features have demonstrated promising outcomes in detecting neurological conditions like epilepsy, stroke, and Alzheimer's disease [36]. By extracting patterns and abnormalities from EEG signals, these classifiers provide objective and quantitative diagnoses, complementing clinician expertise. Integrating machine learning with EEG data offers advantages such as uncovering hidden patterns within complex EEG signals, automating feature extraction, and capturing subtle changes in EEG activity for more sensitive diagnostic models. Large datasets can be utilized, encompassing diverse patient populations and increasing the generalizability of results. Furthermore, real-time or near real-time EEG analysis systems can be developed, enabling rapid and automated assessments, reducing diagnostic delays, and facilitating timely interventions. Combining the power of machine learning with clinical expertise, EEG-based decision support systems have the potential to advance personalized medicine and enhance patient care.

Echo State Networks (ESNs), a highly efficient and performant type of Recurrent Neural Network (RNN), can be used to process temporal data, such as EEG signals [4,34]. ESNs are particularly well-suited for handling sequential data because they have a memory-like property that allows them to retain information from previous time steps. While ESNs and RNNs have shown promise for modeling EEG temporal dynamics due to their memory capabilities, relying solely on individual models may limit robustness. Ensemble methods that combine multiple classifiers have potential to yield more stable predictions. In this work, rather than exclusively using ESNs, we propose an ensemble ESN (E-ESN) method that leverages the strengths of both ESNs and ensemble learning. The E-ESN approach combines predictions from an array of ESN classifiers. This ensemble construction aims to reduce variability and boost performance over a single ESN model. When applied to EEG signals, the recurrent structures within individual ESNs can capture temporal relationships in the data over time. Meanwhile, the ensemble methodology leverages the complementary learning achieved by the diverse array of ESNs. Together, the memory properties of ESNs and integration

through ensemble learning may provide a more robust representation of EEG dynamics compared to relying on isolated RNN architectures.

Following, we present the basic principles of the ESN machine learning technique.

2.2. Echo state networks

Recurrent neural networks (RNNs) are often used for time series prediction tasks like forecasting stock prices, sales data, or weather patterns, but can be difficult to train and require significant computation due to their recurrent architecture [9,22]. Echo state networks (ESNs) address these issues by minimizing training complexity through their unique partially randomized structure [14,24]. ESN learning involves solving a simple linear regression problem to tune the output weights, making training very fast compared to backpropagation through time used to train RNNs [35].

ESNs differ from other recurrent networks in two key ways. First, they have a large reservoir layer, often containing hundreds or thousands of randomly connected sigmoidal neurons, that serves as a rich “reservoir” of temporal features. Second, only the output-reservoir connection weights are learned, while the input-reservoir and recurrent reservoir weights are randomly assigned and remain fixed [33]. The typical ESN architecture includes an input layer, the large reservoir layer, and an output layer (readout) [33] (Fig. 1). The input-reservoir weights are W_{in} , the reservoir connection weights are W , and the reservoir-output weights are W_{out} . An optional W_{back} connection from output to reservoir enables multi-step predictions by feeding predictions back into the network, while without W_{back} , only one-step predictions are made based on the current input and reservoir state [33].

The term “reservoir computing” refers to ESNs and highlights their strengths in achieving high forecasting accuracy comparable to RNNs while being much simpler to train [60].

The state update equations of ESNs are as follows:

$$x(i+1) = f(W_{in} * u(i+1) + W * x(i) + W_{back} * y(i)) \quad (1)$$

$$y(i+1) = g(W_{out} * x(i+1)) \quad (2)$$

Where $x(i)$ is the state of the reservoir at time i , $u(i)$ is the input to the ESN at time i , $y(i)$ is the output of the ESN at time i , W_{in} is the weight matrix between the input layer and the reservoir, W is the weight matrix within the reservoir, W_{back} is the feedback weight matrix from the output layer to the reservoir, f is the activation function of the reservoir neurons, and g is the activation function of the output neurons.

Echo State Networks (ESNs) offer a powerful tool for time series prediction, but their performance is influenced by various critical parameters [24,39]. The spectral radius, acting as a measure of reservoir stability, plays a pivotal role as it must be kept below 1 to ensure the echo state property. Determining the appropriate reservoir scale, which

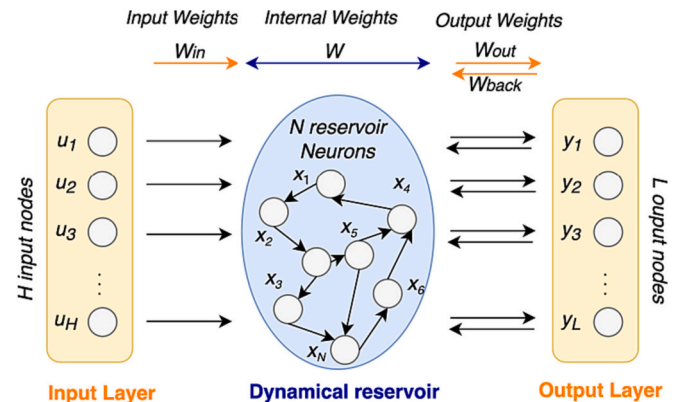


Fig. 1. The Echo State Network architecture [32].

corresponds to the number of neurons in the reservoir, is crucial as it governs the complexity of the task the ESN can effectively handle [19,37]. Additionally, the connectivity rate, reflecting the fraction of connections between neurons in the reservoir, significantly impacts the memory capacity of the ESN [64]. Despite their relative simplicity in training and ability to achieve accurate predictions with a limited number of parameters, ESNs can be sensitive to parameter choices, necessitating meticulous tuning for each specific task at hand [56]. Consequently, achieving optimal parameter settings is vital to unleash the full potential of ESNs and maximize their predictive capabilities in various time series prediction tasks.

Explainability in Echo State Networks (ESNs) is crucial because it provides transparency and interpretability to their predictions and decision-making processes. As machine learning models, standard ESNs can be considered as black boxes, making it challenging to understand how they arrive at their conclusions. In contrast, explainable ESNs incorporate mechanisms that enable users to comprehend the factors contributing to their outputs, making them more trustworthy and accountable in critical applications such as healthcare. This transparency allows domain experts to validate and fine-tune the model, fostering increased confidence and adoption of ESNs in real-world scenarios [63].

2.3. Explainable artificial intelligence

Artificial Intelligence (AI) and Machine Learning (ML) have achieved remarkable accuracy, surpassing human performance [49], and their potential to revolutionize industries, public services, and society is widely acknowledged [65]. However, the inherent “black box” nature of these models raises concerns about their transparency and interpretability [45,51]. The large, complex weights of these models are difficult to comprehend, making it challenging to convey AI decisions to humans, particularly in sensitive domains like healthcare where human life, rights, and privacy are at stake. As AI and ML applications rapidly expand into new domains like digital health, law, transportation, finance, and the military, the demand for transparency and explainability grows [3,9,46]. To address this, Explainable AI (XAI) approaches have emerged, aiming to shed light on the inner workings and decision-making processes of black-box models. These XAI techniques can be broadly categorized as global or local explanations. Global explanations provide insights into the model’s overall decision-making system, while local explanations focus on specific instances or subpopulations. Examples of global XAI techniques include Bayesian Rule Lists (BRL), Generalized Additive Models (GAM), Eli5, Shapley Additive Explanations (SHAP) and the Distillation methodology [31,6,28,59]. On the other hand, local XAI techniques, such as Local Interpretable Model-Agnostic Explanations (LIME), Gradient-weighted Class Activation Mapping (Grad-CAM), and Deep Learning Important Features (DeepLIFT), offer insights into specific instance data [50,38,54,53]. For EEG-based medical Decision Support Systems (DSS), explainability is crucial. Medical professionals need XAI to validate predictions and understand the rationale behind stroke diagnosis decisions. The lack of transparency hinders the adoption of AI models in clinical practice, as physicians require understandable explanations for such critical tasks. Additionally, explainability helps identify and mitigate potential biases that could impact vulnerable groups, fostering trust and acceptance of AI in this high-stakes domain [29]. In fact, XAI offers several noteworthy advantages in the realm of healthcare. Firstly, it enhances transparency and fosters trust in AI-driven systems by granting users visibility into the decision-making processes employed by these systems. Such transparency is pivotal for establishing trust between users and AI systems, thereby facilitating the widespread adoption of AI in healthcare settings. Secondly, XAI empowers users by providing them with insights into the key factors influencing the decisions made by AI systems. This newfound understanding equips users to make more informed decisions regarding their healthcare. Lastly, XAI plays a crucial role in minimizing bias

within AI systems by identifying and mitigating potential biases present in the data or the model. By doing so, XAI ensures that AI systems make fair and impartial decisions, promoting equity and fairness in healthcare practices [41].

3. Multi-level framework for EEG-based medical DSS

EEG-based Medical DSS plays a crucial role in providing accurate and timely assessments and diagnoses. However, the complexity and variability of EEG data pose challenges in developing robust and reliable DSS models. To address these challenges, a multi-level framework has been proposed aiming to enhance the performance and effectiveness of EEG-based DSS. The multi-level framework incorporates a combination of different modules. By utilizing a hierarchical approach, the framework seeks to capture and analyze EEG data at various levels to extract meaningful information and facilitate accurate decision-making. Fig. 2 illustrates the proposed framework. The multi-level framework for enhancing the accuracy and interpretability of EEG-based medical DSS for tasks like stroke prediction consists of the following stages: Data pre-processing is a critical first step in optimizing the quality of the EEG data, enabling ESNs (our adopted machine learning technique) to learn the patterns of brain activity associated with medical conditions. This pre-processing may involve filtering techniques to remove noise, artifact removal to exclude unwanted signals, and data normalization to ensure consistency across recordings. Feature extraction is another important stage that identifies the most informative features in the EEG data, specifically, factors like band powers, allowing the model to focus on relevant stroke biomarkers, allowing ESNs to focus on the most relevant information for medical decision making. During the training stage, the pre-processed and extracted EEG data serve as the foundation for training the ESNs to discern the patterns of brain activity that hold the highest predictiveness for medical conditions, with a specific focus on stroke in our context. This involves training multiple ESNs and then combining their predictions through ensemble learning. This helps improve the accuracy of the system’s decisions. For stroke prediction, the trained Ensemble ESN must then be evaluated on a hold-out test set to predict stroke vs healthy labels. Feature importance must also be analyzed to identify the most influential EEG characteristics. The interpretation stage identifies the patterns of brain activity most predictive of medical outcomes, offering insight into the underlying physiology. This phase may encompass tasks such as visualizing ESN activity and pinpointing the most influential EEG features directly linked to medical decision-making. In the context of stroke prediction, these features are specifically geared toward discerning stroke risk, imparting valuable insights into key physiological indicators. Therefore, combining data pre-processing, feature extraction, ensemble training, and result interpretation aims at enhancing the performance of ESNs for EEG-based medical decision support systems while also providing valuable insights into the physiological basis of various medical conditions. This holistic approach is tailored to ameliorate the learning efficacy in the realm of EEG-based stroke prediction while concurrently yielding valuable clinical perspectives on neurological markers associated with stroke risk.

3.1. Module 1: pre-treatment

EEG data pre-treatment is an essential step in our framework. It involves: (1) EEG data pre-processing to remove noise, artifacts and other unwanted signals from the raw EEG data. This improves the quality of the data for analysis, and (2) feature extraction to identify the most relevant characteristics of the EEG data that can help with medical decision making. A robust feature extraction technique is important to capture the maximum information from the EEG signals. Several feature extraction techniques can be used including time-domain, frequency-domain and non-linear methods. Power spectral density is a well-established frequency-domain approach that decomposes EEG signals

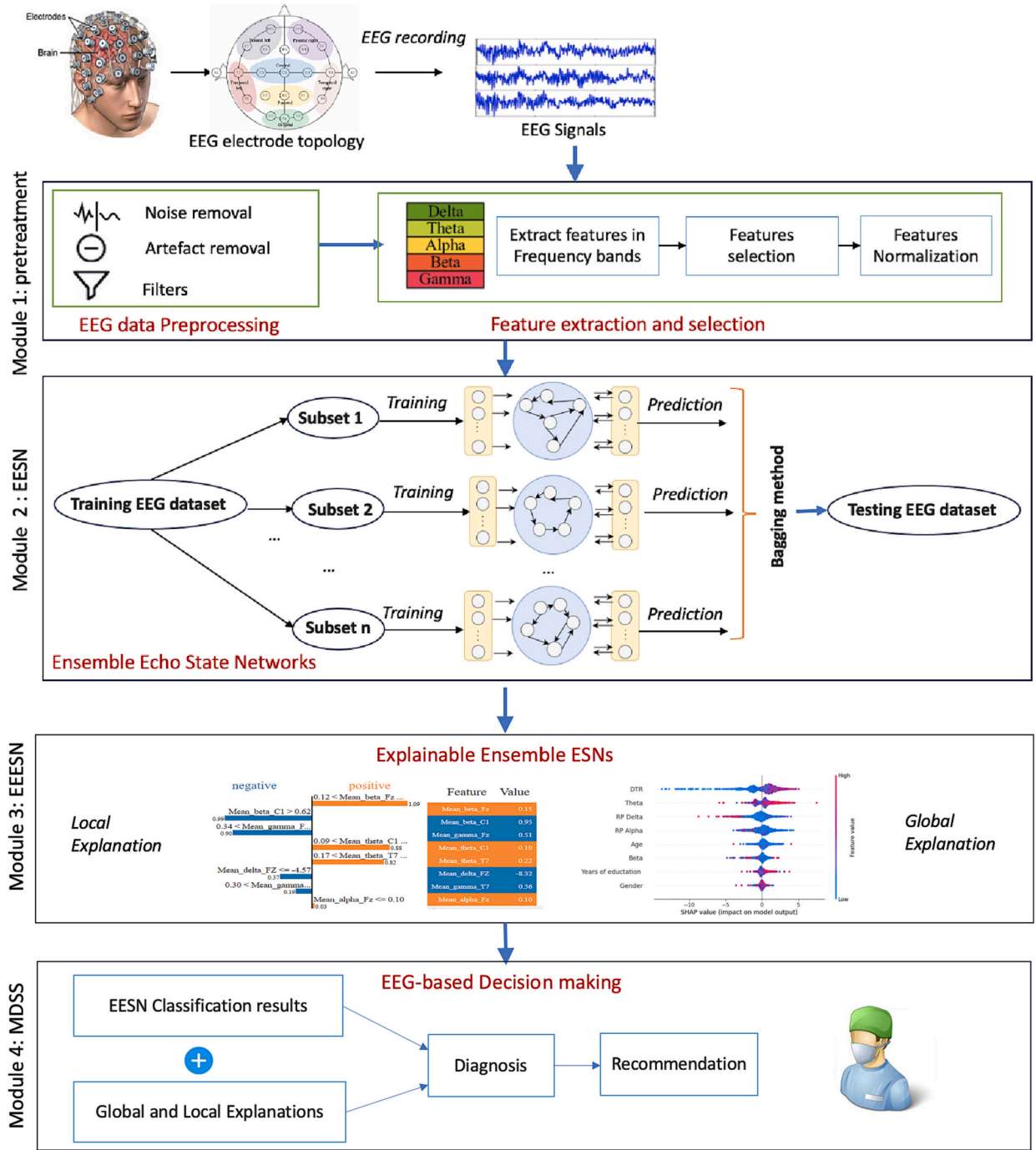


Fig. 2. Multi-Level framework of EEG-based Medical DSS.

into different frequency bands. However, extracting a large number of features can reduce the performance of the model. Therefore, an important step is feature selection to identify a subset of features that are most relevant for the given medical application. This can improve the accuracy and interpretability of the model. In our framework, we propose a novel feature selection algorithm that is designed to identify features from the power spectral density that are most relevant for medical decision making. This optimization is expected to improve the performance of our EEG-based medical decision support system by focusing on the most discriminative features.

The combination of data pre-processing, robust feature extraction using techniques like power spectral density analysis, and optimized feature selection has the potential to improve both the accuracy and

interpretability of EEG-based medical decision support systems. The feature selection algorithm focuses on extracting frequency-band based features from the EEG data rather than mean power in arbitrary frequency ranges. Defining relevant frequency bands can improve selectivity. Calculating the discriminatory power of each frequency band and selecting the most relevant bands can optimize the features for the given medical task.

3.2. Module 2: ensemble Echo State Networks

The second module introduces an ensemble Echo state networks for improved EEG signal classification. In fact, we propose a novel ensemble learning framework for EEG signal classification that combines the

strengths of Echo state networks (ESNs) with the robustness of ensemble learning. Our framework, called ensemble Echo state networks (*E*-ESNs), consists of a pool of ESNs, each with a different reservoir size and connectivity rate. The ESNs are trained on the same dataset, but with different random initializations. The predictions of the ESNs are then combined using the bagging ensemble learning method (cf. Fig. 2) [57]. Bagging is a type of ensemble learning method that creates multiple copies of a model, each trained on a different subset of the training data. The predictions of the models are then aggregated to obtain a final prediction. In our *E*-ESN framework, the ESNs are trained on the same dataset, but with different random initializations. This means that each ESN will be sensitive to different aspects of the temporal data. When the predictions of the ESNs are combined using bagging, this helps to reduce the variance of the ensemble and improve its accuracy. In addition, using bagging can help to prevent overfitting. Overfitting occurs when a model learns the training data too well and is not able to generalize well to new data. By training multiple copies of the model on different subsets of the training data, bagging helps to prevent the models from becoming too similar and overfitting the data.

The state of an ESN at time t is given by the Eq. (3):

$$x_t = \sum_{i=1}^N w_i x_{t-1} + u_t \quad (3)$$

where x_t is the state of the ESN at time t , w_i are the weights of the ESN, N is the number of units in the reservoir, and u_t is the input to the ESN at time t . The output of an ESN is given by the Eq. (4):

$$y_t = \phi(x_t) \quad (4)$$

where ϕ is a nonlinear activation function. The predictions of the ESNs are combined using the bagging method, the predictions of the ESNs are averaged to get a final prediction.

In addition to improving accuracy, *E*-ESNs also have several other advantages over traditional ESNs. First, *E*-ESNs are more robust to overfitting. This is because each ESN is trained on a different subset of the training data, which helps to prevent the models from becoming too similar and overfitting the data. Moreover, *E*-ESNs are more likely to generalize well to new data. This is because the different ESNs are sensitive to different aspects of the data, which means that the ensemble is less likely to be biased toward any particular aspect of the data, and is therefore more likely to generalize well to new data.

3.3. Module 3: explainable ensemble Echo state networks

Our objective is to introduce an innovative Explainable Ensemble Echo State Network (ESN) model designed explicitly for EEG-based medical DSS. In pursuit of this goal, we incorporate two kinds of eXplainable Artificial Intelligence (XAI) techniques: Local Model Explanations and Global Model Explanations. By harnessing the power of our proposed Ensemble ESN model, we aim to achieve not only remarkable accuracy but also comprehensive insights into the intricate facets of EEG signals that underlie the decision-making process. This dual achievement of accuracy and interpretability reinforces the model's credibility, offering healthcare practitioners the means to both trust and comprehend its predictions, thereby advancing the utility and effectiveness of EEG-based medical decision support systems.

3.4. Module 4: EEG-based medical decision-making

In the realm of medical decision-making, the process involves various stages that contribute to informed recommendations. In this regard, we propose a detailed framework specifically designed for the EEG-based Medical Decision-Making Phase. At the initial stage, the framework leverages the classification outcomes of the Ensemble Echo State Network (ESN), which is known for its integrative learning

architecture. These classification outcomes are further enhanced with localized insights using methods like local and global explanation techniques and overarching elucidations. This comprehensive approach affords the classification results a profound level of interpretability. In this module, a careful diagnostic process takes place, where the Ensemble ESN outputs and interpretive explanations are combined to develop a refined understanding of the intricate attributes present in the EEG signals. This enhanced comprehension forms the foundation for precise and astute diagnostic conclusions. This diagnostic phase further elucidates the subtle nuances underlying EEG-based patterns, leading to improved accuracy and reliability in the overall medical decision-making process. This multi-step process culminates in the development of well-informed recommendations. These recommendations draw on the collective synergy of Ensemble ESN classifications, interpretable insights, and diagnostic expertise. By utilizing the outputs of the Ensemble ESN, explanatory revelations, diagnostic acumen, and clinical expertise, the EEG-based Medical Decision-Making Phase serves as a robust conduit, adeptly navigating the complexities of medical scenarios, and ultimately fostering better clinical outcomes and patient-centric care paradigms.

In this section, we suggested that our framework is a comprehensive and robust approach to EEG-based medical decision-making. It addresses all of the key steps in the decision-making process and provides clinicians with the insights they need to make accurate and informed decisions. To validate our framework, we propose to apply it to EEG-based stroke prediction case study.

4. Case study: EEG-based stroke prediction

In this section, we showcase the implementation of our proposed framework in the context of stroke prediction. We rigorously evaluate its performance by presenting the experimental outcomes and conducting a comprehensive assessment to establish its robustness and reliability.

4.1. EEG-based stroke prediction

The incidence of stroke-related diseases is progressively increasing [20] [21], mainly due to changes in environmental factors such as lifestyles, dietary habits, and demanding work cultures [27], [52]. As per the recent report from the World Health Organization (WHO), stroke is the second leading cause of death globally, right after cardiovascular disease [29]. Consequently, the early detection of stroke-related diseases has become a critical requirement for both patients and medical practitioners [48,67]. To optimize stroke diagnosis, the integration of advanced medical instruments and machine learning techniques in computer-aided interventions have become indispensable. Among various diagnostic methods, the use of Electroencephalogram (EEG) stands out as it measures the electrical activity of the brain through electrodes attached to the scalp, identifying abnormalities in brain function [2]. EEG detection offers several advantages, including affordability, simplicity, non-invasiveness, and minimal psychological stress for patients. Research has shown the potential of EEG signals in stroke diagnosis [29]. By leveraging machine learning algorithms, EEG signals can be utilized to classify stroke patients and differentiate them from healthy subjects, or even identify different types of stroke subtypes [44]. The two main types of strokes are ischemic stroke and hemorrhagic stroke. This case study specifically focuses on the classification of stroke patients or control subjects based on EEG data, with the ultimate goal of constructing a prediction model [29]. By employing computer-aided approaches and leveraging the classification capabilities of EEG signals, the aim is to enhance stroke detection and improve patient outcomes.

Recent related works in stroke classification and prediction based on EEG signals have utilized various machine learning techniques to enhance prediction accuracy. For example, Dritsas and Trigka (2022) [13] evaluated machine learning techniques including logistic

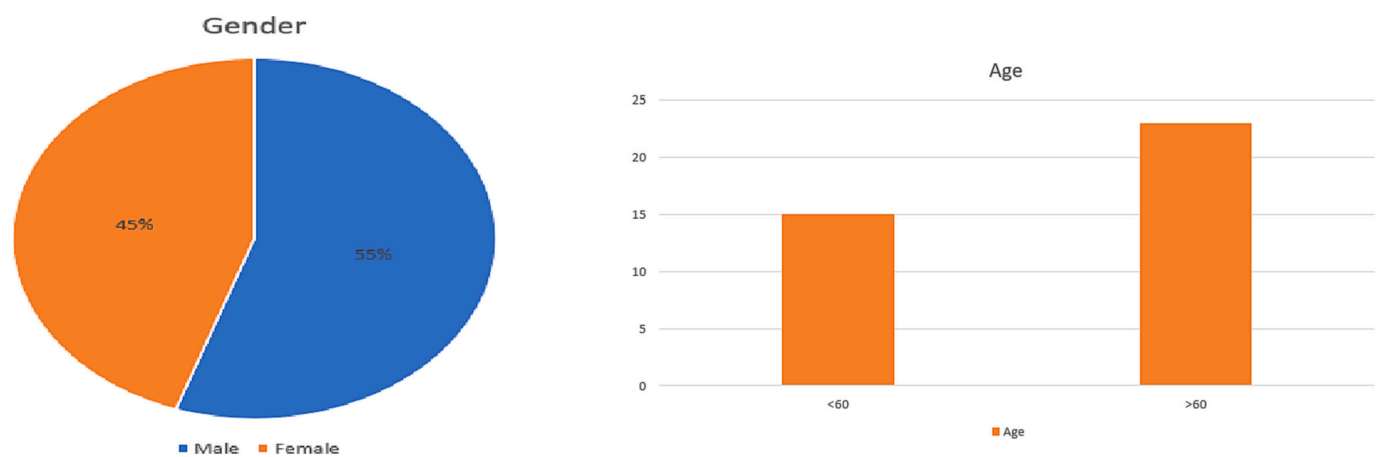


Fig. 3. participants gender and age visualizations.

regression, decision trees, random forest, and gradient boosting machines for stroke risk prediction. They achieved a best accuracy of 80%. Islam et al. (2021) [23] applied machine learning techniques like logistic regression, decision trees, and naïve Bayes for predictive analysis of stroke risk. They achieved a best accuracy of 98%. Kokkotis et al. (2022) [29] proposed an explainable machine learning pipeline using

ensemble methods to address data imbalance for stroke prediction. The authors achieved an accuracy of 73.52%. Darabi et al. (2021) [10] developed a 30-day readmission model for stroke patients using machine learning with an accuracy of 95%. Gupta (2021) [17] compared machine learning algorithms for stroke prediction. They achieved a best accuracy using KNN technique with an accuracy of 95%. Dev et al.

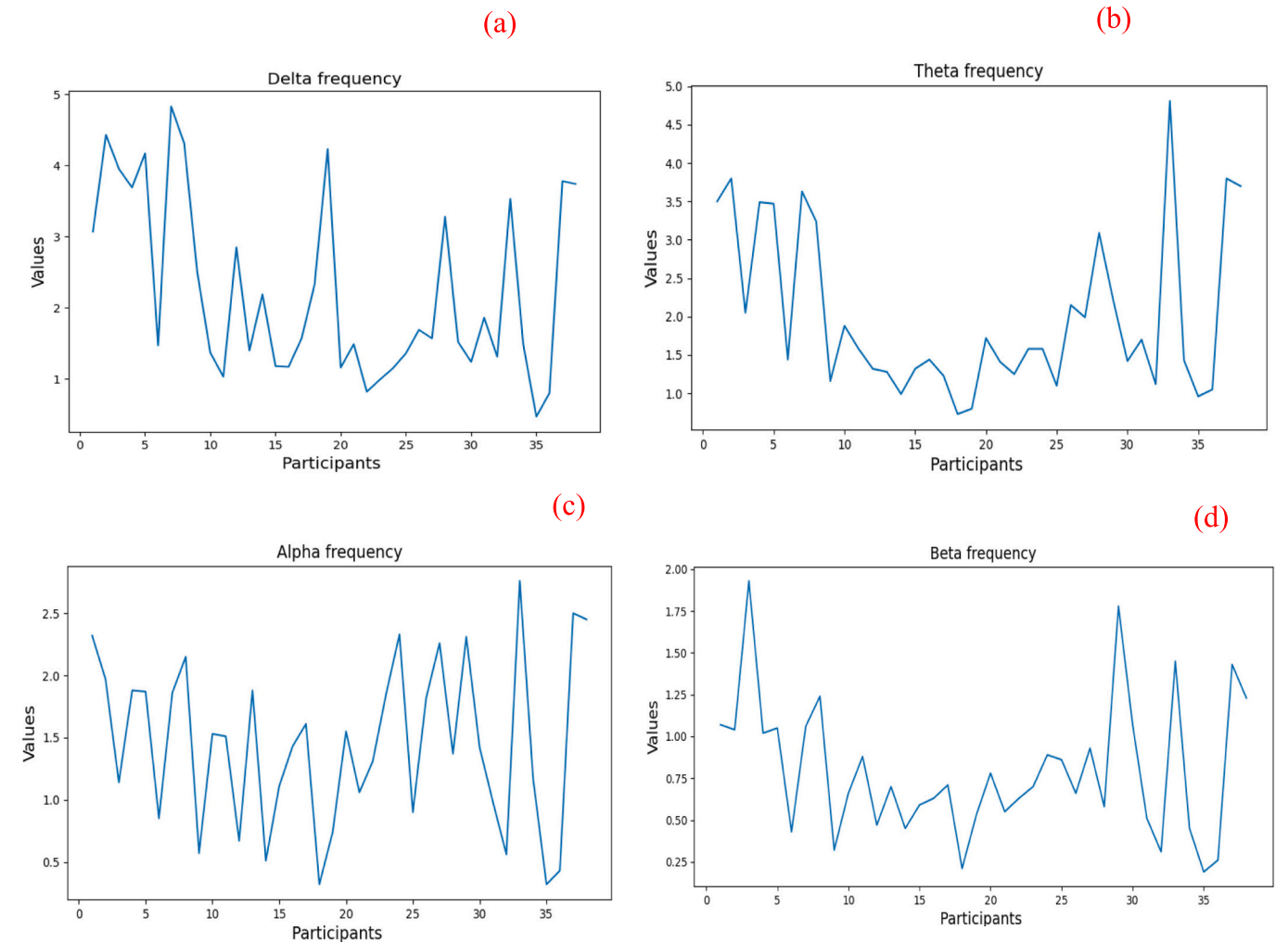


Fig. 4. (a) Delta frequency, (b) Theta frequency, (c) Alpha frequency and (4) Beta frequency.

(2022) [12] presented a predictive analytics approach for stroke prediction using Neural Networks. Their model achieved an accuracy of 77%. Ali (2019) [1] implemented distributed machine learning based on Apache Spark for stroke prediction. He applied four machine learning models: Random Forest, Decision Tree, SVM and Logistic regression with respectively accuracy of 90%, 79%, 77% and 77%. In their research, Liu et al. (2019) [43] introduced a hybrid machine learning method aimed at predicting stroke. The study employed a tool called AutoHPO, which automatically optimized the parameters of a Deep Neural Network (DNN). The DNN model was trained and evaluated using unbalanced physiological data, resulting in an accuracy of 71.6% according to their findings. In a study conducted by Wilkinson et al. (2020) [44], they utilized Random Forest classification trees as a machine learning algorithm. The results showed that this algorithm achieved an accuracy of approximately 76% in distinguishing between stroke and control subjects. The classification was based on the analysis of quantitative EEG and movement variables performed in the study.

Another recent work is of Wang et al. (2023) [62] who proposed a 1D-CNN prediction model for stroke classification based on EEG signal. Their model was able to achieve an accuracy of 90.53%. Mridha et al. (2023) [41] developed automated stroke prediction algorithms that can enable early intervention and improve patient outcomes. By evaluating the effectiveness of a proposed machine learning technique, which achieved an accuracy of almost 91%, and employing explainable methodologies, the study aims to enhance stroke care and treatment through standardized and insightful decision-making.

4.2. Used EEG dataset

This study aimed to identify electrophysiological and clinical factors measured acutely that could predict longer-term post-stroke cognitive outcomes. Cognition is difficult to assess during recovery initially due to issues like fatigue and language problems. The researchers of [2] analyzed EEG data from a single pre-frontal electrode recorded within 72 h of stroke in 24 participants. Statistics on the categorical features of these participants are visible in Fig. 3. Specifically, they measured relative power of different frequency bands and ratios between bands. At around 90 days post-stroke, all participants were given the MoCA to evaluate cognition. Correlation and regression analyses examined relationships between early electrophysiological readings, initial neurological status, other characteristics, and later MoCA scores. Moderate to strong correlations were found for several acute EEG measurements, including delta/theta ratio and power in the theta and delta bands. When combined in a regression model, delta/theta ratio and admission stroke severity best predicted MoCA scores 90 days later, accounting for 75% of the variance. This provides support for the prognostic potential

of early EEG, particularly delta/theta ratio. A single channel recording approach may offer a valid and efficient predictor of post-stroke cognitive function.

While only 24 participants were included, limiting generalizability, studies exploring specific electrophysiological predictors often face data availability challenges. Moreover, the regression analyses established meaningful associations between early EEG markers and subsequent cognition. Successfully predicting MoCA scores to such an extent suggests potentially significant findings despite the modest sample. Nonetheless, larger datasets could enhance result robustness in future work.

4.3. EEG data pre-treatment

4.3.1. Pre-processing

EEG signals are prone to various sources of interference, such as noise, artifacts, and inconsistencies, which can hinder accurate analysis. Hence, it is crucial to preprocess EEG signals and ensure their reliability and suitability for further analysis. In this study, several preprocessing steps were undertaken. They aim to eliminate noise from the EEG signal and identify/remove data affected by movement or muscle activity. This involves amplifying the raw EEG waveform data using a band-pass filter (0.5–30 Hz) to remove noise outside the desired frequency range [47]. Additionally, manual inspection is conducted to identify and mark any data contaminated by movement or muscle artifact, which is then excluded from further processing. Finally, any remaining epochs with amplitudes exceeding $\pm 100 \mu\text{V}$ are removed to eliminate noisy or distorted data unsuitable for further analysis.

4.3.2. Feature selection

The process of feature extraction plays a crucial role in analyzing EEG signals as it enables us to identify and isolate the most significant components of the data. In this particular study, an adapted Power Spectral Density (PSD) analysis was conducted to extract informative frequency-based features from the recorded EEG signals [42]. PSD is a powerful technique for feature extraction in EEG signals. It is well-established, easy to implement, and computationally efficient. However, it is important to be aware of the limitations of PSD before using it. For this, we have adapted it as presented in Algorithm 1. This is achieved by subjecting artifact-free 4-s EEG epochs (with a 1/4 Hz resolution) to Fast Fourier Transforms (FFT) [61]. This conversion from the time-domain EEG signal to the frequency domain enables the analysis of power distribution across different frequency bands. Moreover, relative power (RP) and the delta/alpha ratio (DAR) and delta/theta ratio (DTR) are calculated, as these features have demonstrated correlations with different cognitive states or brain disorders.

Algorithm 1: Adapted PSD-based feature extraction for EEG-based stroke prediction

Input: Artifact-free 4-sec EEG epochs (1/4 Hz resolution)

Steps:

1. Apply Fast Fourier Transforms (FFT) to each EEG epoch to convert from time-domain to frequency-domain.
2. Define frequency bands of interest:
 - Standard bands (delta, theta, alpha, beta)
3. Calculate the power spectral density (PSD) within each band.
4. Extract the relevant features:
 - Relative power (RP) in standard bands
 - Delta/alpha ratio (DAR)
 - Delta/theta ratio (DTR)
5. Normalize feature values to between 0 and 1.

Output : Relative power, DAR and DTR as features to characterize cognitive states or predict stroke outcomes.

Table 1
ESN experimental parameters.

Parameter	Value
Neurons	200
Spectral radius	0.95
Sparsity	0
Noise	0.001

Table 2
Performance metrics of the ESN and the E-ESN.

Metric	Equation	E-ESN Value
Accuracy (Acc)	$Acc = \frac{(TP + TN)}{(TP + TN + FP + FN)} \times 100$	96.50%
Positive Predictive Value (PPV)	$PPV = \frac{(TP)}{(TP + FP)} \times 100$	93.10%
Negative Predictive Value (NPV)	$NPV = \frac{(TN)}{(TN + FN)} \times 100$	90%
Sensitivity (Sen)	$Sen = \frac{(TP)}{(TP + FN)} \times 100$	96.42%
Specificity (Spe)	$Spe = \frac{(TN)}{(TN + FP)} \times 100$	81.81%
F1-score	$F1\text{-score} = 2 \times \frac{(Sen \times PPV)}{(Sen + PPV)}$	94.73%

As a result of our adapted PSD, the Resting State Relative Power (RP) in delta, theta, alpha, and beta frequency bands) provides valuable insights into the distribution of neural activity during the brain's resting state. These ratios offer indications about the mental states and cognitive processes that occur during rest used to predict stroke or control cases. *RP Delta* refers to the ratio of power in the Delta frequency band to the total power across all frequency bands (Fig. 4(a)). Delta waves are associated with deep sleep, relaxation, and unconscious processing. A higher *RP Delta* suggests a greater proportion of slower brainwave activity, possibly indicating deeper relaxation or drowsiness. Similarly, *RP Theta* represents the ratio of power in the Theta frequency band to the total power (Fig. 4(b)). Theta waves are often observed during day-dreaming, light sleep, and creative thinking. A higher *RP Theta* might indicate increased engagement in these mental states during rest. *RP Alpha* represents the ratio of power in the Alpha frequency band (associated with relaxation and a calm, awake state) to the total power (Fig. 4(c)). A higher *RP Alpha* suggests a more relaxed and meditative state during rest. Lastly, *RP Beta* indicates the ratio of power in the Beta frequency band (associated with active thinking, problem-solving, and alertness) to the total power (Fig. 4(d)). A higher *RP Beta* suggests a more alert and focused state, even during rest.

The adapted PSD method selects only the most informative features capable of effectively differentiating stroke patients from healthy individuals. These features are then normalized to a 0 to 1 scale to enable comparative analyses. Therefore, four optimal features are extracted through this approach. The adapted PSD technique enables the rapid isolation of diagnostically crucial components within the complex EEG

signals. Subsequently, these discriminative features can be utilized in accurately classifying stroke patients and non-stroke/control individuals using machine learning algorithms.

In this section, we have described how feature extraction provides valuable insights into the distribution of neural activity during the resting state, allowing us to understand the mental states and cognitive processes that occur. With these feature representations in hand, we can now delve into the machine learning technique to further analyze and interpret the patterns and relationships within the resting state data.

4.4. Ensemble Echo State Networks

To the best of our knowledge, there is a dearth of studies that have specifically examined the utilization of Echo State Networks (ESNs) for the prediction of strokes using EEG data. This knowledge gap serves as a compelling motivation for our research. The ESN is well-suited for capturing the temporal dynamics and patterns in time-series data such as EEG signals [4]. The Ensemble contained seven individual ESNs. Each ESN has its own set of hyperparameters and reservoir configurations. Each ESN model used has a reservoir with 200 neurons, a spectral radius of 0.95 to control the reservoir's dynamics, no sparsity, and a small amount of regularization noise (0.001) to improve generalization (cf. Table 1).

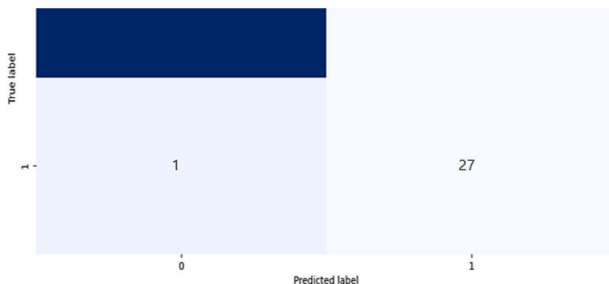
These hyperparameters are critical in configuring the ESN for capturing the temporal patterns in EEG data and effectively distinguishing between stroke patients and healthy individuals based on their EEG patterns. During prediction, the outputs from the seven ESNs are combined through an aggregation method. By training multiple ESNs on the same data but with different random reservoir configurations, the Ensemble approach aims to improve generalizability and avoid overfitting to the training data. The diversity within the Ensemble

Table 3

Accuracy comparison to some related papers' works in terms of accuracy, algorithm and publication year.

Related work	Machine Learning method	Accuracy
Dritsas and Trigka (2022) [13]	NB, LR, KNN, SGD, DT, MLP, RF, Staking	80%
Kokkotis et al. (2022) [29]	LR, RF, KNN, SVM, MLP	73.52%
Darabi et al. (2021) [10]	LR, RF, KNN, SVM, MLP	95%
Islam et al. (2021) [23]	AGB, LR, RF, KNN, SVM, MLP	98%
Gupta (2021) [17]	KNN	95%
Dev et al. (2022) [12]	NN	77%
Ali (2019) [1]	Random Forest	90%
	Decision Tree	79%
	SVM	77%
	Logistic regression	77%
Liu et al. (2019) [43]	DNN	71.6%
Wilkinson et al. (2020) [44]	Random Forest classification trees	76%
Wang et al. (2023) [62]	One-dimensional Convolutional Neural Network	90.53%
Mridha et al. (2023) [41]	LR, RF, KNN, SVC, NB, XGB	91%
Our proposal	E-ESN	96.5%

The accuracy displayed in bold represents the accuracy generated by our system.



Ensemble ESN: **0.9650465824431823**

Fig. 5. The confusion matrix and the generated accuracy.

allows the models to correct each other's errors during the prediction stage. Therefore, the *E*-ESN provides a robust deep learning framework for detecting discriminative EEG biomarkers of stroke using the key frequency-based features extracted through the optimized Power Spectral Density analysis (cf. Algorithm 2).

Conversely, the Specificity (Spe) of 81.81% highlights the model's proficiency in identifying true negative instances, indicating that it correctly classifies a substantial portion of non-stroke cases. The F1-score, which balances precision and recall, underscores the model's overall effectiveness at 94.73%, indicating a balanced performance in capturing both false positives and false negatives. In summation, the

Algorithm 2: Ensemble Echo State Network (E-ESN) for EEG-Based Stroke Prediction

Input:

- Time-series EEG data for stroke prediction.
- Hyperparameters for configuring individual ESNs (e.g., reservoir size, spectral radius, regularization parameter).

Steps:

1. Initialization of Ensemble ESN: Initialize an ensemble of seven individual ESNs, each characterized by unique hyperparameters and reservoir configurations.
2. Configuration of ESN Models:
 - For each ESN within the ensemble:
 - Set the number of neurons in the reservoir (e.g., 200 neurons).
 - Define the spectral radius (e.g., 0.95) governing reservoir dynamics.
 - Specify no sparsity within the reservoir.
 - Incorporate a slight amount of regularization noise (e.g., 0.001) to enhance generalization.
3. Training Phase:
 - Train each ESN instance in the ensemble using the provided EEG data for stroke prediction.
 - Utilize a suitable training algorithm, such as backpropagation through time, to optimize the internal weights of each ESN.
4. Prediction Phase:
 - For each input EEG data sample:
 - Conduct a forward pass through each of the seven ESNs in the ensemble to obtain seven distinct sets of predictions.
 - Employ an ensemble aggregation method, such as weighted averaging or voting, to consolidate predictions from the individual ESNs.
5. Leveraging Ensemble Diversity:
 - Capitalize on the diversity inherent within the ensemble to enhance model generalizability.
 - Exploit the ensemble's capacity to rectify prediction errors through mutual correction among constituent models.

Output:

Composite predictions derived from the seven individual ESNs for each input EEG data sample.

Table 2 presents the performance of the *E*-ESN model, as measured by several metrics derived from the confusion matrix visible in Fig. 5. These metrics include precision (the positive predictive rate), F1 score, sensitivity, specificity and accuracy. Together, they provide a comprehensive view of the performance of the developed ensemble learning algorithm used in the classification task.

Table 2 presents a comprehensive assessment of the performance of the *E*-ESN classification model for predicting stroke. The model's effectiveness is evaluated through a set of key metrics. Notably, the accuracy of the model stands at an impressive 96.50%, highlighting its capability to make correct predictions across the dataset. Moreover, the model demonstrates a robust Positive Predictive Value (PPV) of 93.10%, signifying its accuracy in correctly identifying instances of stroke when predicted as positive. Additionally, the model exhibits a noteworthy Negative Predictive Value (NPV) of 90%, showcasing its proficiency in accurately identifying instances without stroke when predicted as negative. The model's ability to capture true positive instances is measured by its Sensitivity (Sen), which stands at 96.42%, indicating its competence in correctly classifying the majority of actual stroke cases.

model demonstrates a high level of accuracy and showcases notable predictive abilities for stroke detection. While excelling in sensitivity and positive predictive value, the model may benefit from further refinement in capturing instances without stroke. The F1-score encapsulates the model's overall predictive power by harmonizing precision and recall, underscoring its value in providing a comprehensive evaluation of its performance.

To improve the evaluation of our *E*-ESN model and establish its credibility, we conducted a comparative analysis with related works to assess its performance as presented in Table 3.

The presented results in Table 3 offer valuable insights into the related works' accuracy. The diverse algorithms showcased in the studies underscores the concerted effort to develop effective predictive models. Notably, the range of achieved accuracy values indicates both successes and challenges within this domain. Among the studies, Islam et al. (2021) [23] stand out with a remarkable accuracy of 98%, achieved through a skillful ensemble of Adaptive Gradient Boosting, Logistic Regression, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Multi-layer Perceptron. This notable achievement underscores the potential of a well-structured combination of methods for

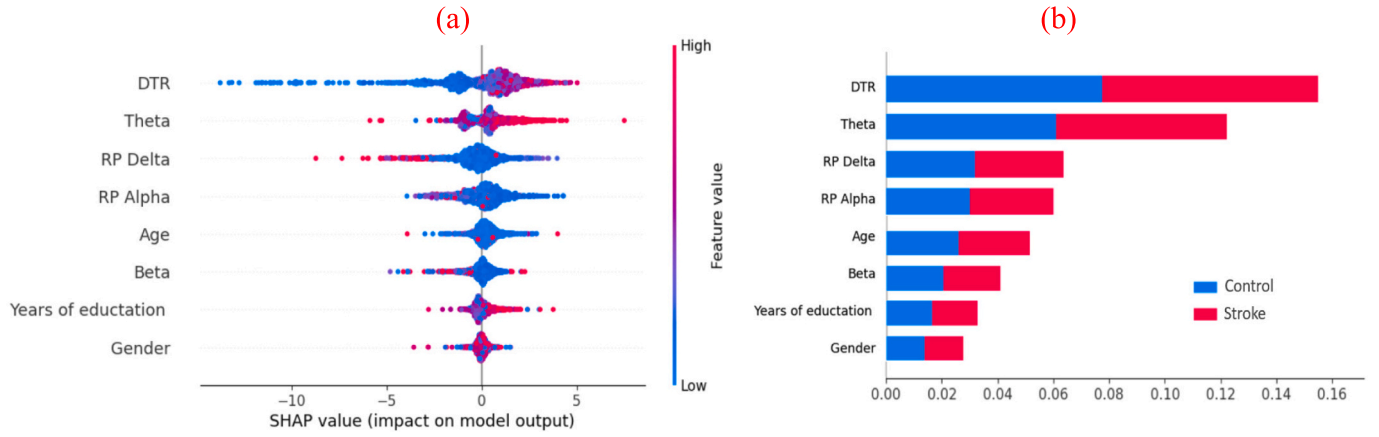


Fig. 6. (a) SHAP value impact on Model output (Each Class), and (b) Average feature impact on model output magnitude.

accurate stroke prediction. Interestingly, our proposal of *E*-ESN, yielding the second-highest accuracy of 96.5%. This result positions *E*-ESN as a promising contender for effective stroke prediction. However, it's worth noting that while *E*-ESN achieves high accuracy, its running time is particularly intriguing, at of 9.35 min. This observation raises the important consideration of computational efficiency alongside predictive performance. Therefore, the accuracy outcomes across different studies emphasizes both the progress and ongoing challenges in stroke prediction. The emergence of *E*-ESN as a strong contender highlights the potential for innovative approaches, while the insight into its running time underscores the need to balance accuracy with computational feasibility in real-world applications.

4.5. Explainable ensemble echo state networks

To better understand our stroke prediction model, we employed explainable artificial intelligence (XAI) tools, i.e., SHAP and LIME. By using these tools, we were able to investigate how the *E*-ESN model make predictions and how each input attribute influences the prediction. By identifying the key features that significantly contribute to the prediction, we can reduce the feature space and save time during model training while improving accuracy. Additionally, we can focus on collecting more data points that include these important features, which will further enhance the model's prediction performance.

4.5.1. Global model explanation using SHAP

The SHAP (SHapley Additive exPlanations) is an XAI (Explainable Artificial Intelligence) technique that aims to provide interpretability for

machine learning models. It is based on the concept of Shapley values from cooperative game theory. SHAP assigns a value to each feature in a prediction, indicating its contribution toward the prediction outcome. It considers all possible feature combinations and quantifies the impact of each feature by averaging across these combinations. The resulting SHAP values provide insights into the importance and influence of individual features on the model's predictions. By using SHAP, users can gain a better understanding of how different features affect the model's decision-making process. It helps identify which features have a positive or negative impact on predictions and allows for the identification of potential biases or influential factors in the model's output.

It is important to understand the effect of each predictor variable on the complex model's results. We calculate the mean Shapley Additive Explanations (SHAP) values from the *E*-ESN classifier to determine the common feature impacts. Fig. 6 illustrates the top eight factors with the greatest effects: DTR, Theta, RP Delta, RP Alpha, age, beta, years of education, and gender. These important features align with existing domain knowledge from experts and literature regarding their theoretical explanations.

The *E*-ESN classifier's average feature importance estimates for each class (0 and 1) are depicted in Fig. 6(a) and (b). As shown by the Shapley values on the x-axis, the average feature value contributes very little to the total potential collaborations of the output. A negative, zero, or positive contribution is represented when the Shapley value is lower than 0, equal to 0, or higher than 0, respectively. The y-axis orders the attributes by decreasing relevance, while the right longitudinal axis indicates the range of characteristic values from low to high.

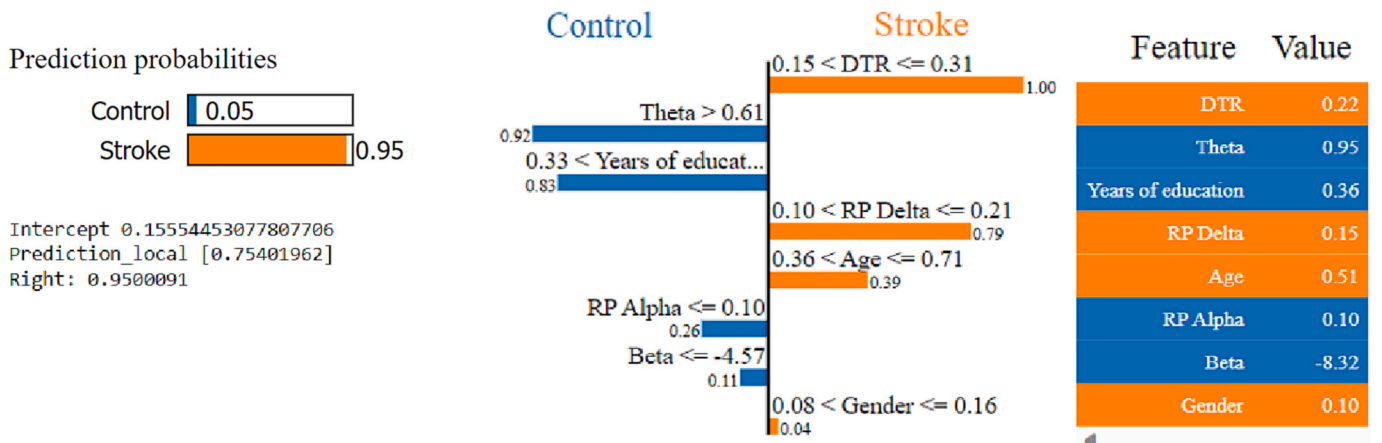


Fig. 7. LIME explanation.

Stroke Detection

Patient information

Patient data

First name: ***** Last name: ***** Age: >60 Gender: Male Handedness: Right Years of education: 15

EEG data

Alpha: 1.14 Beta: 1.93 Theta: 2.05 Delta: 3.95 RP Alpha: 0.13 RP Beta: 0.21 RP Theta: 0.23 RP Delta: 0.44 DTR: 1.93 DAR: 3.46

Explainable decision-making

Prediction

Diagnosis: 95 % **Stroke risk**

Interpretation

LIME

SHAP

Recommendation

☐ Magnetic Resonance Imaging ☐ D-Dimer test

☐ Complete Blood Count test ☐ Lipid Profile test

☐ Basic Metabolic Panel test ☐ Coagulation Profile test

Cancel Confirm

Fig. 8. User Interface of the EEG-based stroke prediction DSS.

4.5.2. Local model explanation using LIME

LIME (Local Interpretable Model-Agnostic Explanations) is an approach designed to offer understandable explanations for any machine learning model, the E-ESN in our specific context. Instead of directly explaining the complex model, LIME approximates it with a simpler, more interpretable model that produces similar predictions. It focuses on explaining individual predictions rather than the entire model, generating local explanations that accurately reflect how the original model behaves for a particular instance. Importantly, LIME achieves this without requiring access to the internal parameters of the model. By producing visual explanations, such as bar charts and feature value tables, LIME reveals the most influential features for a prediction, providing valuable insights into the decision-making process of the model for a given example.

Fig. 7 illustrates two examples of explanations from the dataset. In the first instance, the initial predicted class is “Control” (i.e. healthy patient), and the important predictive features with the greatest impact on this classification can be seen from the feature values. Similarly, in the second instance the predicted class is “Stroke”, and the explanation identifies the key features that informed this prediction. This provides physicians with transparency into the computer-aided diagnosis to build understanding and trust in the model’s predictions. Interpreting E-ESN model locally can be challenging. However, LIME does an excellent job of explaining these predictions by investigating how the values of different features either increase or decrease the model’s certainty in its result. We can see some features have a strong positive or negative influence on the location of the forecast. The average feature impact calculates the average effect of each feature across all instances on the model’s predicted output (control or stroke). It quantifies how much the prediction would change if a single feature value was altered while keeping all other feature values the same. This provides insight into the general predictive importance of different features in the model.

4.6. Medical decision-making for stroke prediction

The developed EEG-based medical decision support systems integrates categorical and EEG data to analyze it using E-ESN, predict

stroke risk and provide recommendations for further diagnostic tests. The user-friendly interface provided in Fig. 8 is accessible by the physician. It includes patient information, EEG data, predictions, interpretations, and recommendations for further analysis or confirmation.

The interface presents patient information relevant to assessing stroke risk. It includes demographic data like name, age, gender, and years of education. It also provides EEG readings measuring brain activity levels (e.g. alpha, beta waves). Based on this data, the model has predicted there is a 95% risk of stroke for this patient. This result suggests that the system has analyzed the data and made a high-confidence prediction. The interpretation section mentions the availability of LIME and SHAP buttons, indicating the use of local and global interpretable explanations to explain the model’s predictions (cf. Figs. 7 and 8). The “LIME” button likely shows feature importance via a LINE (local interpretation) plot. The “SHAP” button presumably displays SHAP (Shapley additive explanations) values to further explain the key drivers behind the prediction. Based on the case of this patient, the recommendation to the physician suggests specific diagnostic tests, such as Magnetic Resonance Imaging (MRI), Complete Blood Count (CBC) test, Basic Metabolic Panel (BMP) test, Coagulation Profile test, Lipid Profile test, and D-Dimer test. This could help validate or refine the initial prediction. These tests are likely recommended based on the predicted stroke risk and the need for further evaluation. The physician can choose between these recommendations, confirm or cancel them.

5. Discussion

In this paper we introduce a multi-level framework for EEG-based medical DSS. The framework addresses the challenges posed by complex and variable EEG data by incorporating a hierarchical approach, encompassing data pre-processing, feature extraction, ensemble learning, result interpretation, and decision-making. The primary aim is to enhance the accuracy and interpretability of EEG-based DSS through strategic module integration. The framework consists of four modules. Module 1 focuses on data pre-treatment, which includes noise and artifact removal, as well as feature extraction. The framework

introduces a novel feature selection algorithm to extract the most relevant frequency-band based features. This enhances both model accuracy and interpretability. Module 2 introduces Ensemble Echo State Networks (E-ESNs), a robust classification approach bolstered by ensemble learning. E-ESNs consist of multiple ESNs trained on the same data but with diverse initializations, reducing model variance and mitigating overfitting. This integration of E-ESNs leverages their sensitivity to different temporal data aspects, ultimately enhancing model robustness and generalizability. Module 3 amalgamates eXplainable Artificial Intelligence (XAI) techniques, specifically Local and Global Model Explanations. By embedding these within the E-ESN model, the framework not only achieves high accuracy but also delivers comprehensive insights into the intricate aspects of EEG signals driving decisions. This combination of accuracy and interpretability enhances model credibility, enabling healthcare practitioners to trust and comprehend predictions, thereby advancing EEG-based medical DSS. Finally, Module 4 culminates in EEG-based Medical Decision-Making. It leverages E-ESN classifications and interpretive insights, fusing them with diagnostic expertise to refine understanding and reach astute conclusions. This phase bridges the complexities of medical scenarios, fostering improved clinical outcomes and patient-centric care paradigms.

Therefore, the proposed framework establishes a robust methodology for EEG-based medical decision-making. By addressing each key step of the process and offering clinicians the insights needed for informed decisions, the framework lays the groundwork for enhanced clinical practices. The upcoming EEG-based stroke prediction case study will serve as a validation of this comprehensive approach. From a practical standpoint, the proposed framework was applied to a comprehensive case study that focuses on utilizing EEG data and machine learning techniques for stroke prediction. The study highlights the significance of EEG in stroke prediction, emphasizing its non-invasiveness and simplicity, which make it a valuable tool for early detection. It underscores the importance of integrating advanced medical instruments and machine learning techniques for accurate and timely diagnosis. The study emphasizes the crucial role of data pre-processing in ensuring the reliability of EEG data. Specifically, it highlights the importance of noise removal and artifact exclusion. Feature extraction using Power Spectral Density (PSD) analysis is employed to capture relevant frequency-based features from EEG signals. The choice of features and preprocessing techniques significantly impact the performance of predictive models. Then, it introduces a novel approach called Ensemble Echo State Networks (E-ESN), which combines the strengths of ESNs and ensemble learning. The study demonstrates the potential of the E-ESN model with an accuracy of 96.5%, making it competitive with other related works. In this work, the need for model interpretability in medical decision-making is addressed. It applies explainable artificial intelligence (XAI) tools such as SHAP and LIME to provide insights into the E-ESN model's predictions. This transparency enhances trust and comprehension among healthcare practitioners, making the model more actionable and reliable. Finally, using a specific user interface, the healthcare practitioners (i.e., physician) are able to assess stroke risk in patients and make informed decisions. The findings of the study suggest that EEG-based stroke prediction has a promising future in enhancing clinical decision-making and patient outcomes. The integration of feature extraction methods, ensemble learning, and explainability tools offers a comprehensive framework that can be extended to other medical domains for improved diagnostics and treatment planning. Therefore, the practical implications of this work extend to both research and healthcare practice, promising advancements in early stroke detection and patient-centric care paradigms.

However, it is important to note that stroke severity exists on a spectrum, ranging from mild or transient ischemic attacks to severe cases with significant neurological impairment and major infarction. Detecting abnormalities in EEG patterns may vary depending on stroke location and severity, with more subtle or early-stage strokes potentially leading to subtle EEG changes that are challenging to discern with

certainty. The study's limitation lies in its inclusion of participants who underwent EEG recording within 72 h of a stroke, primarily capturing moderate to severe cases based on included severity scores. Acutely occurring milder strokes may have produced more ambiguous EEG findings, limiting the framework's capability to predict outcomes for minor stroke cases solely based on early EEG data. Moreover, factors such as stroke location could influence the manifestation and interpretability of EEG signatures, potentially affecting predictive model performance. Another limitation is that the framework does not evaluate patients' ability to perform tasks, such as producing a signature, which could provide additional insights into subtle neurological changes indicating stroke risk or impairment. As shown by Das Chakladar et al. [11], combining behavioral assessments like signatures with EEG in a multimodal model using Siamese Neural Networks improved verification performance over unimodal approaches.

To address these limitations, future work will aim to incorporate user verification techniques. Building upon the multimodal Siamese network approach of Das Chakladar et al. [11], we plan to collect additional behavioral data from patients, such as signature production ability, to fuse with EEG signals. This will help detect subtle impairment in mild stroke cases and improve verification of patient identity. Verified patient identity and behavioral markers could augment the EEG-based Medical Decision Support System, enhancing its accuracy and generalizability across the spectrum of stroke severity. Larger and more diverse patient cohorts will also be recruited to thoroughly explore the impact of stroke severity, location and other clinical factors on early EEG and behavioral findings as predictors of outcomes. This will allow validating the framework for a wider range of real-world stroke presentations. Addressing these limitations through planned extensions, such as multimodal user verification, could advance the system toward clinical deployment for personalized stroke risk assessment and management."

6. Conclusion

Modern healthcare has been revolutionized by medical decision support systems, enabling precise care delivery. Electroencephalogram (EEG) data integration within these systems has become pivotal, offering valuable insights for brain-computer interfaces, disease prediction, and cognitive assessment. Leveraging advanced techniques and machine learning, EEG-based Medical Decision Support Systems empower timely and informed decisions, elevating healthcare standards. Our proposed multi-level framework for EEG-based systems includes pre-processing, adapted feature selection, ensemble Echo State Networks, AI-driven explanations, and decision support. The main contributions of this work are: (1) proposing a multi-level framework for EEG-based medical DSS that addresses challenges of complex and variable EEG data through a hierarchical approach, (2) introducing a novel feature selection algorithm to extract the most relevant frequency-band based features from EEG data, enhancing both model accuracy and interpretability, (3) proposing an Ensemble Echo State Networks (E-ESN) classification approach that leverages ensemble learning to reduce model variance and mitigate overfitting, improving robustness and generalizability, (4) Embedding explainable AI (XAI) techniques within the E-ESN model to not only achieve high accuracy but also provide comprehensive model explanations and credibility for healthcare practitioner, and (5) bridging machine learning classifications and interpretive insights with diagnostic expertise through a medical decision-making module, fostering improved clinical outcomes.

In fact, this research was applied to stroke prediction. The study primarily focused on identifying frequency-based features with discriminative properties, achieved through an adapted Power Spectral Density (PSD) analysis of EEG recordings. The E-ESN model enhanced the accuracy of stroke classification to be equal to 96.5%. Notably, XAI tools, namely SHAP and LIME, were employed to gain deeper insights into the decision-making process of the E-ESN model and decision support to assist physicians take appropriate decisions. These XAI

techniques elucidated the crucial role played by specific EEG frequency bands, such as theta, alpha, beta and delta, recorded at distinct electrode locations, in contributing to accurate stroke prediction. This enhanced understanding not only enhances model transparency but also guides future data collection endeavours, prioritizing significant features. The performance comparison with related works demonstrates the superiority of the proposed E-ESN model over the majority of prevalent machine learning techniques used for stroke prediction. This achievement signifies a substantial advancement in the domain of stroke diagnosis, underscoring the potential of deep learning frameworks in EEG data classification. The implications of this research hold significant promise for healthcare practitioners, offering a more efficient and precise system of stroke diagnosis, facilitating timely interventions, and improving patient outcomes.

To enhance the generalizability and accuracy of our model, we plan to expand our dataset to include diverse cohorts of stroke patients and healthy individuals across varying demographics and subtypes. Integrating EEG data with complementary neuroimaging approaches like fMRI and PET can provide deeper insights into the neural correlates of stroke. We aim to develop real-time EEG-based systems that can enable early stroke detection and timely interventions. We also plan to incorporate user verification techniques by collecting behavioral data from patients, such as assessing signature production ability. Fusing these behavioral measures with EEG signals using approaches like multimodal Siamese networks could help detect subtle impairment across the spectrum of stroke severity. Verified patient identity and behavioral markers have the potential to enhance accuracy and generalizability of our EEG-based decision support system. We believe these efforts will significantly advance AI-driven diagnostics and care in neurology through more comprehensive assessment of patients. Further optimization of feature selection, online learning, and transfer learning techniques will improve model performance and adaptability. As AI-based diagnostics gain prominence, addressing ethical data privacy and security considerations is imperative. Building on our stroke prediction methodology's success, we see opportunities to extend this approach to other neurological disorders like epilepsy, Alzheimer's, and Parkinson's disease.

CRedit authorship contribution statement

Samar Bouazizi: Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – original draft. **Hela Ltfi:** Supervision, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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